Uyuni 2026.08

安装和升级指南

章 1. 前言

Installation, Deployment and Upgrade + Uyuni 2026.08

本指南提供了部署、升级和管理 Uyuni 服务器及代理的详尽分步说明。

指南分为以下几个章节：

- **要求**：概述确保安装流程顺畅进行的必要硬件、软件和网络前提条件。
- **部署与安装**：指导您将 Uyuni 部署为容器并完成初始配置。
- **升级与迁移**：详细说明升级和迁移 Uyuni 的过程，同时最大限度减少停机时间。
- **基本的服务器管理**：涵盖基本的服务器操作，帮助您高效上手使用 Uyuni。

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章 2. 要求

2.1. 一般要求

下表指定了服务器和代理的最低要求。

请勿使用 NFS 存储数据，因为它不支持 SELinux 文件标记。

2.1.1. 服务器要求

表格 1. x86-64 体系结构的服务器要求

Software and Hardware	Details	Recommendation
Tumbleweed	Clean installation, up-to-date	Tumbleweed
CPU	-	Minimum 4 dedicated 64-bit CPU cores (x86-64)
RAM	Test or Base Installation	Minimum 16 GB
	Production Server	Minimum 32 GB
Disk Space	/ (root directory)	Minimum 40 GB
	/var/lib/containers/storage/volumes	Minimum 150 GB (depending on the number of products)
	/var/lib/containers/storage/volumes/var-pgsql	Minimum 50 GB
	/var/lib/containers/storage/volumes/var-spacewalk	Minimum storage required: 100 GB (this will be verified by the implemented check) * 每个 SUSE 产品和软件包中心 50 GB 360 GB for each Red Hat product
	/var/lib/containers/storage/volumes/var-cache	Minimum 10 GB. Add 100 MB per SUSE product, 1 GB per Red Hat or other product. Double the space if the server is an ISS Master.
	Swap space	3 GB

2.1.2. 代理要求

表格 2. 代理要求

Software and Hardware	Details	Recommendation
Tumbleweed	Clean installation, up-to-date	Tumbleweed
CPU		Minimum 2 dedicated 64-bit CPU cores
RAM	Test Server	Minimum 2 GB
	Production Server	Minimum 8 GB
Disk Space	/ (root directory)	Minimum 40 GB
	/var/lib/containers/storage/volumes	Minimum 100 GB

Uyuni Proxy caches packages in the **/var/lib/containers/storage/volumes/uyuni-proxy-squid-cache/** directory. If there is not enough space available in **/var/lib/containers/storage/volumes/uyuni-proxy-squid-cache/**, the proxy will remove old, unused packages and replace them with newer packages.

鉴于这种行为：

- The larger **/var/lib/containers/storage/volumes/uyuni-proxy-squid-cache/** directory is on the proxy, the less traffic there will be between it and the Uyuni Server.
- By making the **/var/lib/containers/storage/volumes/uyuni-proxy-squid-cache/** directory on the proxy the same size as **/var/lib/containers/storage/volumes/var-spacewalk/** on the Uyuni Server, you avoid a large amount of traffic after the first synchronization.
- The **/var/lib/containers/storage/volumes/uyuni-proxy-squid-cache/** directory can be small on the Uyuni Server compared to the proxy. 有关大小估算的指导，请参见 服务器要求 一节。

2.2. 网络要求

本节详细说明 Uyuni 的网络和端口要求。

> IP 转发将通过容器化安装来实现。这意味着 Uyuni 服务器和代理将充当路由器。此行为由 podman 直接完成。如果禁用 IP 转发，podman 容器将不会运行。
>
> 您可以考虑根据您的策略实现 Uyuni 环境的网络隔离。
>
> 有关详细信息，请参见 https://www.suse.com/support/kb/doc/?id=000020166。

2.2.1. 完全限定的域名 (FQDN)

Uyuni 服务器必须正确解析其 FQDN。如果无法解析 FQDN，可能会导致许多不同的组件出现严重问题。

有关配置主机名和 DNS 的详细信息，请参见 https://documentation.suse.com/sles/15-SP7/html/SLES-all/cha-network.html#sec-network-yast-change-host。

2.2.2. 主机名和 IP 地址

为确保 Uyuni 域名可由其客户端解析，服务器和客户端计算机都必须连接到一台正常工作的 DNS 服务器。还需要确保正确配置反向查找。

有关设置 DNS 服务器的详细信息，请参见 https://documentation.suse.com/sles/15-SP7/html/SLES-all/cha-dns.html。

2.2.3. 重新启用路由器通告

When the Uyuni is installed using **mgradm install** or **mgrpxy install**, it sets up Podman which enables IPv4 and IPv6 forwarding. This is needed for communication from the outside of the container.

However, if your system previously had **/proc/sys/net/ipv6/conf/eth0/accept_ra** set to **1**, it will stop using router advertisements. As a result, the routes are no longer obtained via router advertisements and the default IPv6 route is missing.

要恢复 IPv6 路由的正常功能，请根据以下场景执行相应步骤：

- 服务器和代理基于 15 SP7（无 Network Manager）
- server and proxy are based on SL Micro 6.2 (with Network manager)

过程：无 Network Manager 时重新启用路由器通告

1. Create a file in **/etc/sysctl.d**, for example **99-ipv6-ras.conf**.
2. 向文件中添加以下参数和值：

```
net.ipv6.conf.eth0.accept_ra = 2
```

3. 重引导。

Network management is not working when you have no wicked.

过程：有 Network Manager 时重新启用路由器通告

1. List your connections with **nmcli connection show**.
2. Create or modify the file **/etc/NetworkManager/system-connections/<name of connection>.nmconnection** to add this setting:

```
[ipv6]
addr-gen-mode=eui64
```

3. 重引导。

4. 文件示例如下：

```
[connection]
id=Wired connection 1
type=ethernet
interface-name=eth0

[ethernet]

[ipv4]
dns-priority=20
method=auto

[ipv6]
addr-gen-mode=eui64
method=auto
```

2.2.4. 在 HTTP 或 HTTPS OSI 七层代理后部署

有些环境强制要求通过 HTTP 或 HTTPS 代理（如 Squid 服务器或类似服务器）访问互联网。要在此类配置下允许 Uyuni 服务器访问互联网，需进行以下配置。

过程：配置 HTTP 或 HTTPS OSI 七层代理

1. For operating system internet access, modify **/etc/sysconfig/proxy** according to your needs:

```
PROXY_ENABLED="no"
HTTP_PROXY=""
HTTPS_PROXY=""
NO_PROXY="localhost, 127.0.0.1"
```

2. For **Podman** container internet access, modify **/etc/systemd/system/uyuni-server.service.d/custom.conf** according to your needs. For example, set:

```
[Service]
Environment=TZ=Europe/Berlin
Environment="PODMAN_EXTRA_ARGS="
Environment="https_proxy=user:password@http://192.168.10.1:3128"
```

3. For Java application internet access, modify **/etc/rhn/rhn.conf** according to your needs. On the container host, execute **mgrctl term** to open a command line inside the server container:

   a. Modify **/etc/rhn/rhn.conf** according to your needs. For example, set:

```
# Use proxy FQDN, or FQDN:port
server.satellite.http_proxy =
server.satellite.http_proxy_username =
server.satellite.http_proxy_password =
```

```
# no_proxy is a comma separated list
server.satellite.no_proxy =
```

4. 在容器主机上，重启服务器以强制应用新配置：

```
systemctl restart uyuni-server.service
```

2.2.5. 隔离的部署

如果您在内部网络中操作，无法访问 SUSE Customer Center，可以使用 **Installation and Upgrade Guide › Container Deployment › 隔离的部署**。

在生产环境中，Uyuni 服务器和客户端始终应使用防火墙。有关所需端口的完整列表，请参见 **Installation and Upgrade Guide › Network Requirements › Ports**。

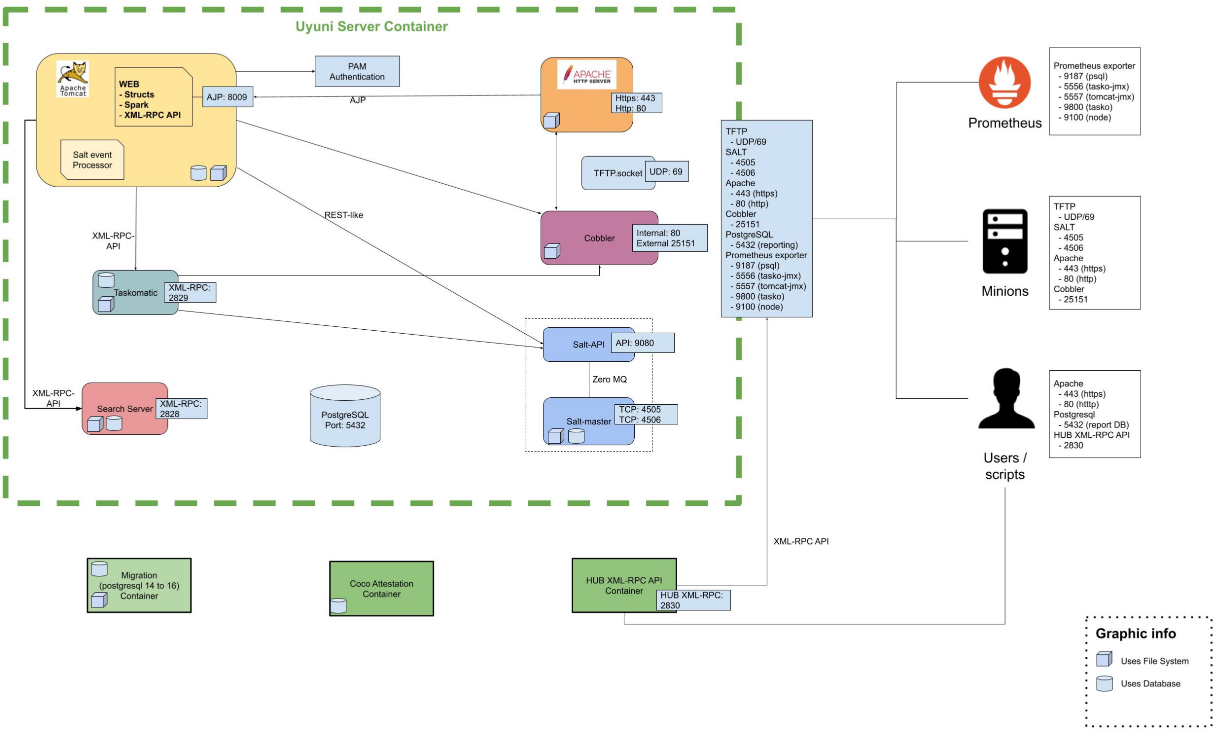
2.2.6. 所需的网络端口

本节提供了 Uyuni 中各种通信使用的端口的综合列表。

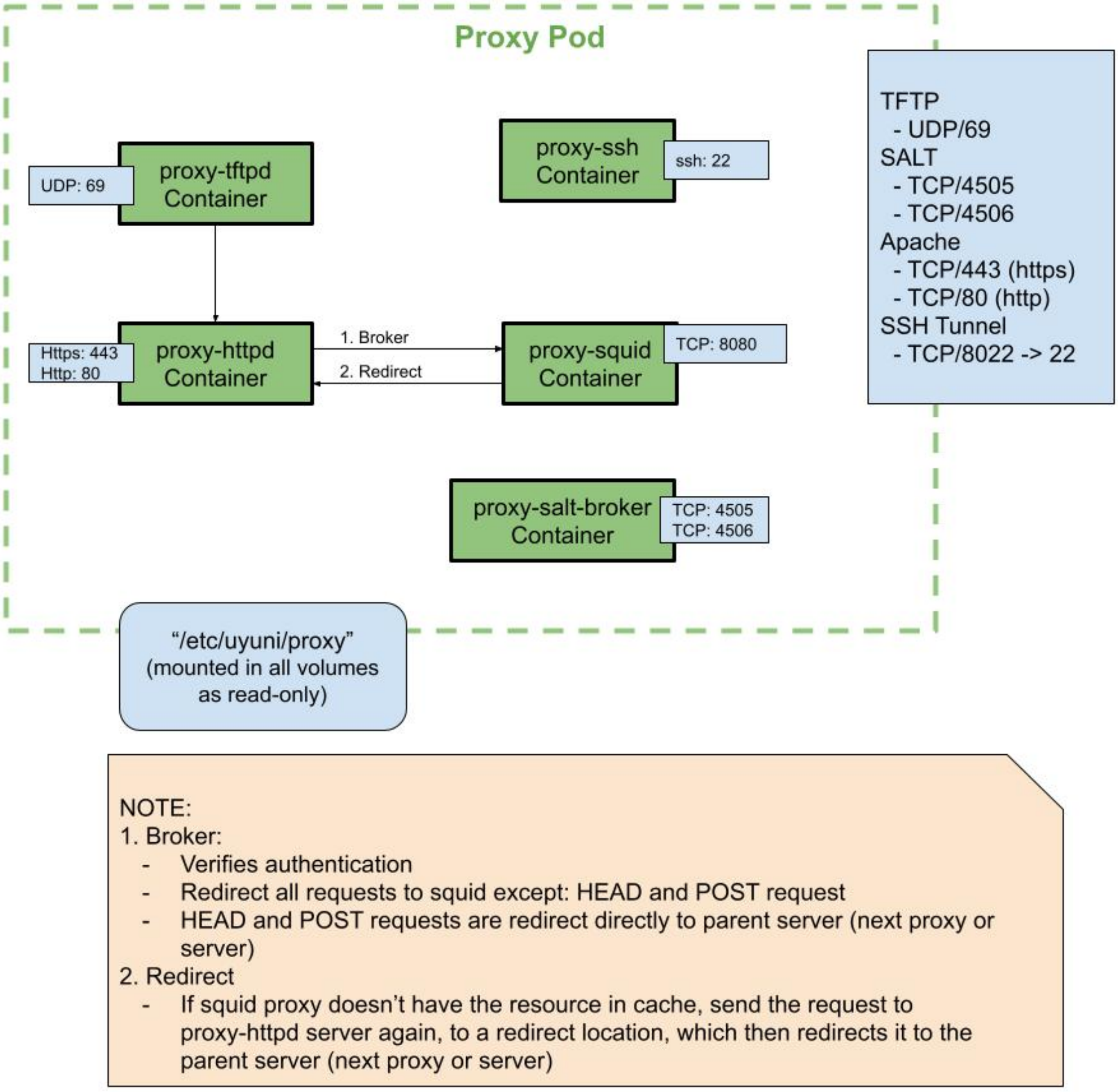
您不需要打开所有这些端口。某些端口只有在您使用需要这些端口的服务时才需打开。

2.2.6.1. 概述

2.2.6.1.1. 服务器



2.2.6.1.2. 代理



2.2.6.2. 外部入站服务器端口

必须打开外部入站端口，以在 Uyuni 服务器上配置防火墙用于防范未经授权访问服务器。

打开这些端口将允许外部网络流量访问 Uyuni 服务器。

表格 3. Uyuni Server 的外部端口要求

Port number	Protocol	Used By	Notes
67	TCP/UDP	DHCP	Required only if clients are requesting IP addresses from the server.
69	TCP/UDP	TFTP	Required if server is used as a PXE server for automated client installation.

Port number	Protocol	Used By	Notes
80	TCP	HTTP	Required temporarily for some bootstrap repositories and automated installations.
443	TCP	HTTPS	Serves the Web UI, client, and server and proxy (**tftpsync**) requests.
4505	TCP	salt	Required to accept communication requests from clients. The client initiates the connection, and it stays open to receive commands from the Salt master.
4506	TCP	salt	Required to accept communication requests from clients. The client initiates the connection, and it stays open to report results back to the Salt master.
5432	TCP	PostgreSQL	Required to access the reporting database.
5556	TCP	Prometheus	Required for scraping Taskomatic JMX metrics.
5557	TCP	Prometheus	Required for scraping Tomcat JMX metrics.
9100	TCP	Prometheus	Required for scraping Node exporter metrics.
9187	TCP	Prometheus	Required for scraping PostgreSQL metrics.
9800	TCP	Prometheus	Required for scraping Taskomatic metrics.

2.2.6.3. 外部出站服务器端口

必须打开外部出站端口，以在 Uyuni 服务器上配置防火墙用于限制服务器可访问的内容。

打开这些端口将允许来自 Uyuni 服务器的网络流量与外部服务通信。

表格 4. Uyuni Server 的外部端口要求

Port number	Protocol	Used By	Notes
80	TCP	HTTP	Required for SUSE Customer Center. Port 80 is not used to serve the Web UI.
443	TCP	HTTPS	Required for SUSE Customer Center.

2.2.6.4. 内部服务器端口

Internal ports are used internally by the Uyuni Server. Internal ports are only accessible from **localhost**.

大多数情况下无需调整这些端口。

表格 5. Uyuni Server 的内部端口要求

端口号	备注
2828	Satellite-search API，由 Tomcat 和 Taskomatic 中的 RHN 应用程序使用。
2829	Taskomatic API，由 Tomcat 中的 RHN 应用程序使用。
8005	Tomcat 关机端口。
8009	Tomcat 到 Apache HTTPD (AJP)。
8080	Tomcat 到 Apache HTTPD (HTTP)。
9080	Salt-API，由 Tomcat 和 Taskomatic 中的 RHN 应用程序使用。
32000	与运行 Taskomatic 和 satellite-search 的 Java 虚拟机 (JVM) 建立 TCP 连接时使用的端口。

32768 和更高的端口用作临时端口。这些端口往往用于接收 TCP 连接。收到 TCP 连接请求后，发送方将选择其中一个临时端口号来与目标端口进行匹配。

可使用以下命令来确定哪些端口是临时端口：

```
cat /proc/sys/net/ipv4/ip_local_port_range
```

2.2.6.5. 外部入站代理端口

必须打开外部入站端口，以在 Uyuni Proxy 上配置防火墙用于防范未经授权访问代理。

打开这些端口将允许外部网络流量访问 Uyuni Proxy。

表格 6. Uyuni Proxy 的外部端口要求

Port number	Protocol	Used By	Notes
22			Only required if the user wants to manage the proxy host with Salt SSH.
67	TCP/UDP	DHCP	Required only if clients are requesting IP addresses from the server.
69	TCP/UDP	TFTP	Required if the server is used as a PXE server for automated client installation.
443	TCP	HTTPS	Web UI, client, and server and proxy (**tftpsync**) requests.
4505	TCP	salt	Required to accept communication requests from clients. The client initiates the connection, and it stays open to receive commands from the Salt master.

Port number	Protocol	Used By	Notes
4506	TCP	salt	Required to accept communication requests from clients. The client initiates the connection, and it stays open to report results back to the Salt master.
8022			Required for ssh-push and ssh-push-tunnel contact methods. Clients connected to the proxy initiate check in on the server and hop through to clients.

2.2.6.6. 外部出站代理端口

必须打开外部出站端口，以在 Uyuni Proxy 上配置防火墙用于限制代理可访问的内容。

打开这些端口将允许来自 Uyuni Proxy 的网络流量与外部服务通信。

表格 7. Uyuni Proxy 的外部端口要求

端口号	协议	使用方	备注
80			用于访问服务器。
443	TCP	HTTPS	SUSE Customer Center 需要此端口。
4505	TCP	Salt	直接或通过代理连接到 Salt 主控端时需要此端口。
4506	TCP	Salt	直接或通过代理连接到 Salt 主控端时需要此端口。

2.2.6.7. 外部客户端端口

必须打开外部客户端端口，以在 Uyuni 服务器及其客户端之间配置防火墙。

大多数情况下无需调整这些端口。

表格 8. Uyuni 客户端的外部端口要求

端口号	方向	协议	备注
22	入站	SSH	使用 ssh-push 和 ssh-push-tunnel 联系方法时需要此端口。
80	出站		用于访问服务器或代理。
443	出站		用于访问服务器或代理。
4505	出站	TCP	直接或通过代理连接到 Salt 主控端时需要此端口。
4506	出站	TCP	直接或通过代理连接到 Salt 主控端时需要此端口。
9090	出站	TCP	Prometheus 用户界面需要此端口。

端口号	方向	协议	备注
9093	出站	TCP	Prometheus 警报管理器需要此端口。
9100	出站	TCP	Prometheus 节点导出器需要此端口。
9117	出站	TCP	Prometheus Apache 导出器需要此端口。
9187	出站	TCP	Prometheus PostgreSQL 需要此端口。

2.2.6.8. 所需的 URL

Uyuni 必须能够访问某些 URL 才能注册客户端和执行更新。大多数情况下，允许访问以下 URL 便已足够：

- **scc.suse.com**
- **updates.suse.com**
- **installer-updates.suse.com**
- **registry.suse.com**
- **registry-storage.suse.com**
- **opensuse.org**

In order to perform confidential computing attestation for IBM, the system needs to be to be able to autonomously download the certificates and the certificate-revocation lists. Therefore the following addresses must be reachable, even behind a HTTP proxy:

- **www.ibm.com/support/resourcelink/api/content/public/*** (this includes IBM certificates and IBM CRL distribution points)
- **crl3.digicert.com/** and **crl4.digicert.com** (this includes CRL distribution points for **DigiCertCA.crt**, since the chain must be checked for`` the CRLs)

此外，您可能需要访问以下非 SUSE 产品的 URL：

- **download.nvidia.com**
- **public.dhe.ibm.com**
- **nu.novell.com**

You can find additional details on whitelisting the specified URLs and their associated IP addresses in this article: Accessing SUSE Customer Center and SUSE registry behind a firewall and/or through a proxy.

如果您正在使用非 SUSE 客户端，则还可能需要允许访问为这些操作系统提供特定软件包的其他服务器。例如，如果您使用的是 Ubuntu 客户端，则需要能够访问 Ubuntu 服务器。

有关为非 SUSE 客户端排查防火墙访问权限问题的详细信息，请参见 **Administration Guide › Troubleshooting › Tshoot Firewalls**。

2.3. 公有云要求

本节介绍在公有云基础结构上安装 Uyuni 所要满足的要求。我们已在 Amazon EC2、Google Compute Engine 和 Microsoft Azure 上对这些指令进行过测试，不过它们进行一定修改后在其他提供商的云服务上也应能正常工作。

在开始之前，请注意以下一些事项：

- Uyuni 设置过程执行正向确认的反向 DNS 查找。此操作必须成功，设置过程才能完成，并且 Uyuni 才能按预期方式运行。请务必在设置 Uyuni 之前执行主机名和 IP 配置。
- Uyuni Server 和 Proxy 实例需在适当的网络配置中运行，该网络配置可让您控制 DNS 项，但无法通过因特网自由访问。
- 在此网络配置中必须提供 DNS 解析：**hostname -f** 必须返回完全限定的域名 (FQDN)。
- DNS 解析对于连接客户端也很重要。
- DNS 取决于所选的云框架。有关详细说明，请参见云提供商文档。
- 我们建议将软件储存库、服务器数据库和代理 squid 缓存存储在外部虚拟磁盘上。这可以防止在实例意外终止时丢失数据。本节包含有关设置外部虚拟磁盘的说明。

2.3.1. 网络要求

在公有云上使用 Uyuni 时，必须使用受限制的网络。我们建议使用带有适当防火墙设置的 VPN 专用子网。只能允许指定 IP 范围内的计算机访问该实例。

在公有云上运行 Uyuni 意味着需要实施强大的安全措施。限制、过滤、监控并审计对实例的访问至关重要。SUSE 强烈建议不要配置全球均可访问但缺少充足边界安全保护的 Uyuni 实例。

要访问 Uyuni Web UI，请在配置网络访问控制时允许 HTTPS。这将允许您访问 Uyuni Web UI。

In EC2 and Azure, create a new security group, and add inbound and outbound rules for HTTPS. In GCE, check the **Allow HTTPS traffic** box under the **Firewall** section.

2.3.2. 准备存储卷

我们建议将 Uyuni 的储存库和数据库存储在不同于根卷的存储设备上。这有助于避免丢失数据，有时还可以提高性能。

Uyuni 容器利用默认存储位置。应在部署之前为自定义存储配置这些位置。有关详细信息，请参见**Installation and Upgrade Guide › Container Management › 永久性容器卷**

不要使用逻辑卷管理 (LVM) 进行公有云安装。

用于存储储存库的磁盘大小取决于您要使用 Uyuni 管理的发行套件和通道数目。挂接虚拟磁盘时，它们将作为 Unix 设备节点显示在实例中。设备节点的名称因提供商及所选实例类型而异。

确保 Uyuni 服务器的根卷大小不少于 100 GB。如果可能，请另外添加一个 500 GB 或以上大小的存储磁盘，并选择 SSD 存储类型。当您的实例启动时，Uyuni 服务器的云映像会使用脚本来指派这个单独的卷。

启动实例后，您便可登录 Uyuni 服务器，并使用以下命令查找所有可用的存储设备：

```
hwinfo --disk | grep -E "Device File:"
```

If you are not sure which device to choose, use the **lsblk** command to see the name and size of each device. Choose the name that matches with the size of the virtual disk you are looking for.

You can set up the external disk with the **mgr-storage-server** command. This creates an XFS partition mounted at **/manager_storage** and uses it as the location for the database and repositories:

```
/usr/bin/mgr-storage-server <devicename>
```

章 3. 部署和安装

3.1. 安装 Uyuni 服务器

部署 Uyuni 服务器的场景多种多样。

3.1.1. Uyuni Server Deployment on openSUSE Tumbleweed

3.1.1.1. 部署准备工作

In this section, you will gain expertise in setting up and deploying a Uyuni Server. The process encompasses the installation of **Podman**, **Uyuni container utilities**, deployment, and then initiating interaction with the container through **mgrctl**.

> This section assumes you have already configured an openSUSE Tumbleweed host server, whether it is running on a physical machine or within a virtual environment.
>
> https://download.opensuse.org/tumbleweed/

3.1.1.2. 容器主机一般要求

有关一般要求，请参见**Installation and Upgrade Guide › 一般要求**。

An openSUSE Tumbleweed server should be installed from installation media.

https://download.opensuse.org/tumbleweed/

下面介绍此过程。

3.1.1.3. 容器主机要求

有关 CPU、RAM 和存储要求，请参见**Installation and Upgrade Guide › 硬件要求**。

> 为了保证客户端能够解析 FQDN 域名，容器化服务器和主机都必须连接到正常运行的 DNS 服务器。此外，必须确保反向查找的配置正确。

3.1.1.4. 安装用于容器的 Uyuni 工具

Procedure: Installing Uyuni Tools on openSUSE Tumbleweed

1. On your local host, open a terminal window and log in.
2. Add the following repository to your openSUSE Tumbleweed server. You might need to use **sudo** for the following commands.

```
zypper ar
https://download.opensuse.org/repositories/systemsmanagement:/Uyuni:/Stable/ima
ges/repo/Uyuni-Server-POOL-$(arch)-Media1/ uyuni-server-stable
```

3. Refresh the repository list and import the key:

   ```
   zypper ref
   ```

   When prompted, trust and import the new repository GPG key.

4. 安装容器工具：

   ```
   zypper in mgradm mgrctl mgradm-bash-completion mgrctl-bash-completion uyuni-
   storage-setup-server
   ```

有关 Uyuni 容器实用程序的详细信息，请参见 Uyuni 容器实用程序。

3.1.1.5. 配置自定义永久性存储

This step is optional. However, if custom persistent storage is required for your infrastructure, use the **mgr-storage-server** tool.

For more information, see **mgr-storage-server --help**. This tool simplifies creating the container storage and database volumes.

如下所示使用命令：

```
mgr-storage-server <storage-disk-device> [<database-disk-device>]
```

3.1.1.6. 例如：

mgr-storage-server /dev/nvme1n1 /dev/nvme2n1

```
[NOTE]
====
This command will create the persistent storage volumes at
`/var/lib/containers/storage/volumes`.

有关详细信息，请参见 xref:installation-and-upgrade:container-management/persistent-
container-volumes.adoc[]。
====


== Install and Enable Podman

Before deploying {productname}, you must install and enable `podman`.

.Procedure: Installing and Enabling Podman
[role=procedure]
----

. Log in as root and install:
+
[source, shell]
```

zypper in podman

```
. Start and enable the Podman service:
+
[source, shell]
```

systemctl enable --now podman.service

```
----


== 使用 Podman 部署 Uyuni 容器

=== `mgradm` Overview

{productname} is deployed as a container using the `mgradm` tool. There are two methods of
deploying a {productname} server as a container. In this section we will focus on basic
container deployment.

有关使用自定义配置文件进行部署的信息，请参见 xref:installation-and-upgrade:container-
management/mgradm-yaml-custom-configuration.adoc[]。

For additional information, you can explore further by running `mgradm --help` from the
command line.


:leveloffset: +2

[NOTE]
====
{productname} server hosts that are hardened for security may restrict execution of files
from the `/tmp` folder. In such cases, as a workaround, export the `TMPDIR` environment
variable to another existing path before running `mgradm`.

例如:

[source, shell]
```

export TMPDIR=/path/to/other/tmp

```
在 {productname} 的后续更新中，相关工具将进行优化，届时无需再使用此临时解决方案。
====

:leveloffset: 3

.过程: 使用 Podman 部署 Uyuni 容器
. 在终端中以 sudo 或 root 用户身份运行以下命令。
+
[source, shell]
```

sudo mgradm install

```
+
[IMPORTANT]
====
必须以 sudo 或 root 用户身份部署容器。如果您遗漏此步骤，终端中将显示以下错误。
```

```
[source, shell]
```

INF Setting up uyuni network 9:58AM INF Enabling system service 9:58AM FTL Failed to open /etc/systemd/system/uyuni-server.service for writing error="open /etc/systemd/system/uyuni-server.service: permission denied"

```
====

. 等待部署完成。
. 打开浏览器并访问您的服务器 FQDN。

//In this section you learned how to deploy an {productname} Server container.



=== 永久性卷

许多用户希望指定其永久性卷的位置。

[NOTE]
====
If you are just testing out {productname} you do not need to specify these volumes.
`mgradm` will setup the correct volumes by default.

通常只需为较大规模的生产部署指定卷位置。
====

By default `podman` stores its volumes in `/var/lib/containers/storage/volumes/`.

You can provide custom storage for the volumes by mounting disks on this path or the
expected volume path inside it such as: `/var/lib/containers/storage/volumes/var-
spacewalk`. This is especially important for the database and package mirrors.

有关容器中所有永久性卷的列表，请参见：

* xref:installation-and-upgrade:container-management/persistent-container-volumes.adoc[]
* xref:administration:troubleshooting/tshoot-container-full-disk.adoc[]



:leveloffset!:
:leveloffset: +3

= {productname} 服务器隔离的部署
:revdate: 2025-07-25
:page-revdate: {revdate}

== 什么是隔离的部署？


隔离部署是指设置和操作与不安全网络（尤其是互联网）隔离的任何联网系统。这种部署通常用于军事
设施、金融系统、关键基础架构等高安全性环境，以及处理敏感数据，因而必须防范其受到外部威胁的
任何位置。

You can easily pull container images using `Podman` or `Docker` on a machine with internet
access.

.过程：拉取映像
. Pull the desired images, then save the images as a `tar` archive. For example:

+
.Podman
```

podman pull registry.opensuse.org/uyuni/server:latest registry.opensuse.org/uyuni/server-postgresql:latest podman save -m --output images.tar registry.opensuse.org/uyuni/server:latest registry.opensuse.org/uyuni/server-postgresql:latest

```
+
.Docker
```

docker pull registry.opensuse.org/uyuni/server:latest registry.opensuse.org/uyuni/server-postgresql:latest docker save --output images.tar registry.opensuse.org/uyuni/server:latest registry.opensuse.org/uyuni/server-postgresql:latest

```
+
. Transfer the resulting `images.tar` to the Server container host and load it using the
following command:
+
.加载服务器映像
```

podman load -i images.tar

```
== 在隔离环境中获取 {salt} 公式的容器映像

部分公式（如 Bind 和 DHCP
(Kea)）也使用容器。如果您计划在隔离环境中使用这些公式，需要拉取其映像、将映像保存为归档，
然后加载到 {productname} 代理或其他受管系统中。

映像可从 `registry.opensuse.org` 获取。

.过程：为隔离环境获取公式映像
. 在能够访问互联网的系统上，拉取所需映像。例如：
+
[source, shell]
```

podman pull registry.opensuse.org/opensuse/bind:latest podman pull registry.opensuse.org/opensuse/kea:latest

```
. 将映像保存为 TAR 归档：
+
[source, shell]
```

podman save -o formula-images.tar registry.opensuse.org/opensuse/bind:latest registry.opensuse.org/opensuse/kea:latest

```
. 将 `formula-images.tar` 文件传输至隔离的系统。
. 在隔离的系统上加载映像：
+
[source, shell]
```

podman load -i formula-images.tar

```
=== Deploy {productname} on {opensuse} {tumbleweed}

{productname} 还在 RPM 软件包中提供了可在系统上安装的全部所需容器映像。
```

```
[NOTE]
====
用户应在内部网络上提供所需的 RPM。这可以通过使用第二个 {productname}
服务器或任意类型的镜像来完成。
====

.Procedure: Install {productname} on {opensuse} {tumbleweed} in air-gapped environment
. Install {opensuse} {tumbleweed}.
. 更新系统。
. 安装工具软件包和映像软件包（将 $ARCH$ 替换为适当的体系结构）：

+

[source, shell]
```

zypper install mgradm* mgrctl* uyuni-server*-image* uyuni-server-proxy-tftpd-image*

```
The TFTPD image is now the same than on the proxy. If TFTPD is disabled on the server this
image is not needed.

+

. Deploy {productname} with `mgradm`. In an Air-gapped environment you may want to use the
option `--pullPolicy Never`.


For more detailed information about installing {productname} Server on {opensuse}
{tumbleweed}, see xref:container-deployment/uyuni/server-deployment-uyuni.adoc[Server
Deployment].

To upgrade {productname} Server, apply available host updates with `transactional-update`,
reboot if updates were applied, then run `mgradm upgrade`. For air-gapped installations,
upgrade the container RPM packages before running `mgradm upgrade`.


:leveloffset: 3
:leveloffset: +2

[[installation-proxy]]
= 安装 {productname} 代理
:revdate: 2025-02-19
:page-revdate: {revdate}

// **This file is needed to link generically to proxy installation**

部署 {productname} 代理的场景多种多样。所有这些场景都假定您已成功部署 {productname}
{productnumber} 服务器。

:leveloffset: 3
:leveloffset: +3

[[installation-proxy-containers]]
= {productname} Proxy Deployment on {opensuse} {tumbleweed}
:revdate: 2025-10-08
:page-revdate: {revdate}


本指南概述了 {productname} {productnumber} 代理的部署过程。本指南假定您已成功部署
{productname} {productnumber} 服务器。要成功完成部署，请执行以下操作：


.核对清单：代理部署
```

```
. 查看硬件要求。
. Install {opensuse} {tumbleweed} on a bare-metal machine.
. 将代理作为 {salt} 受控端进行引导。
. 生成代理配置。
. 将服务器中的代理配置传输到代理
. 使用代理配置将 {salt} 受控端作为代理注册到 {productname}。


.代理容器主机支持的操作系统
[NOTE]
====
The supported operating system for the container host is {opensuse} {tumbleweed}.

容器主机:: 容器主机是配备了容器引擎（例如
Podman）的服务器，可用于管理和部署容器。这些容器包含应用程序及其必备组件（例如库），但不包
含完整的操作系统，因此体量很小。此设置可确保应用程序能够在不同环境中以一致的方式运行。容器
主机为这些容器提供必要的资源，例如 CPU、内存和存储。
====



== 代理的硬件要求


下表列出了部署 {productname} 代理所要满足的硬件要求。

[cols="1,3,2", options="header"]
.代理硬件要求
|===

| Hardware
| Details
| Recommendation

| CPU
| {x86_64}, {arm}
| Minimum 2 dedicated 64-bit CPU cores

| RAM
| Minimum
| 2 GB

|
| Recommended
| 8 GB

| Disk Space
| `/` (root directory)
| Minimum 40 GB

|
| `/var/lib/containers/storage/volumes`
| Minimum 100 GB, Storage requirements should be calculated for the number of ISO
distribution images, containers, and bootstrap repositories you will use.

|===



== 容器主机一般要求

有关一般要求，请参见xref:installation-and-upgrade:general-requirements.adoc[一般要求]。

An {opensuse} {tumbleweed} server should be installed from installation media. This
procedure is described below.
```

```
[[installation-proxy-containers-requirements]]
== 容器主机要求

有关 CPU、RAM 和存储要求，请参见xref:installation-and-upgrade:hardware-
requirements.adoc[硬件要求]。


[IMPORTANT]
====
为了保证客户端能够解析 FQDN 域名，容器化服务器和主机都必须连接到正常运行的 DNS
服务器。此外，必须确保反向查找的配置正确。
====


== 配置自定义永久性存储

This step is optional. However, if custom persistent storage is required for your
infrastructure, use the `mgr-storage-proxy` tool.

For more information, see `mgr-storage-proxy --help`. This tool simplifies creating the
container storage and Squid cache volumes.

如下所示使用命令：
```

mgr-storage-proxy <存储磁盘设备>

```
例如：
```

mgr-storage-proxy /dev/nvme1n1

```
[NOTE]
====
This command will create the persistent storage volumes at
`/var/lib/containers/storage/volumes`.

有关详细信息，请参见

* xref:installation-and-upgrade:container-management/persistent-container-volumes.adoc[]
* xref:administration:troubleshooting/tshoot-container-full-disk.adoc[]
====


== 将代理主机作为受控端进行引导

.任务：引导代理主机
. 选择menu:系统[引导]。
. 填写代理主机的相关字段。
. 从下拉列表中选择上一步骤中创建的激活密钥。
. 单击 btn:[+ 引导]。
. 等待引导过程成功完成。检查 menu:Salt[] 菜单，确认 {salt} 受控端密钥已列出并已接受。
. 重引导代理主机。
. 从menu:系统[]列表中选择主机，并在所有事件完成后再次触发重引导以完成初始配置。


.任务：更新代理主机
. 从menu:系统[]列表中选择主机，并应用所有补丁以将其更新。
. 重引导代理主机。
```

```
// FIXME 2024-12-10, ke: use the snippet also here (see MLM equiv)
[[proxy-setup-containers-generate-config]]
== 生成代理配置

{productname} 代理的配置归档由 {productname}
服务器生成。每个附加代理都需要自身的配置归档。

[IMPORTANT]
====
在生成此代理配置之前，必须将 {productname} 代理的容器主机作为 Salt 受控端注册到
{productname} 服务器。
====


您将执行以下任务：

.过程：
. 生成代理配置文件。
. 将配置传输到代理。
. Start the Proxy with the `mgrpxy` command.

[[proc-proxy-containers-setup-webui]]
.任务：使用 Web UI 生成代理容器配置

. 在 {webui} 中，导航到menu:系统[代理配置]，然后填写所需数据：
. In the `Proxy FQDN` field type fully qualified domain name for the proxy.
. In the `Parent FQDN` field type fully qualified domain name for the {productname} Server
or another {productname} Proxy.
. In the `Proxy SSH port` field type SSH port on which SSH service is listening on
{productname} Proxy. Recommended is to keep default 8022.
. In the `Max Squid cache size [MB]` field type maximal allowed size for Squid cache.
Typically this should be at most 60% of available storage for the containers.
In the `SSL certificate` selection list choose if new server certificate should be
generated for {productname} Proxy or an existing one should be used. You can consider
generated certificates as {productname} builtin (self signed) certificates.
+
然后根据所做的选择提供用于生成新证书的签名 CA
证书的路径，或者要用作代理证书的现有证书及其密钥的路径。
+
The CA certificates generated on the server are stored in the
`/var/lib/containers/storage/volumes/root/_data/ssl-build` directory.
+
有关现有或自定义证书的详细信息以及企业和中间证书的概念，请参见 xref:administration:ssl-
certs-imported.adoc[]。

. 单击 btn:[生成] 以在 {productname} 服务器中注册新代理
FQDN，并生成包含容器主机细节的配置归档。
. 片刻之后，系统会显示文件可供下载。请将此文件保存在本地。


[[proxy-deploy-containers-transfer-config]]
== 传输代理配置

{webui} 将生成配置归档。需要在代理容器主机上提供此归档。


.任务：复制代理配置

. Copy the files to the Proxy host:
+
```

scp config.tar.gz <代理 FQDN>:/root

```
. 使用以下命令安装代理：
+
```

mgrpxy install config.tar.gz

```
[[proxy-deploy-containers-transfer-start]]
== 启动 {productname} {productnumber} 代理

Container can now be started with the `mgrpxy` command:

[[proc-install-containers-setup-start]]
.任务：启动代理并检查状态

. 调用以下命令启动代理：
+
```

mgrpxy start

```
. 调用以下命令检查容器状态：
+
```

mgrpxy status

```
+

Five {productname} Proxy containers should be present and should be part of the `proxy-
pod` container pod:

* proxy-salt-broker
* proxy-httpd
* proxy-tftpd
* proxy-squid
* proxy-ssh

== 安装用于容器的 Uyuni 工具


.Procedure: Installing Uyuni Tools on {opensuse} {tumbleweed}
[role=procedure]


----
. Go to menu:System Details[Software > Packages > Install].
. Install the `uyuni-tools` package.
----


有关 Uyuni 容器实用程序的详细信息，请参见
link:https://build.opensuse.org/repositories/systemsmanagement:Uyuni:Stable:ContainerUtils
[Uyuni 容器实用程序]。

=== 为服务使用自定义容器映像

By default, the {productname} Proxy suite is set to use the same image version and
registry path for each of its services. However, it is possible to override the default
values for a specific service using the install parameters ending with `-tag` and `-
image`.

例如，可以按如下方式使用此命令：
```

mgrpxy install --httpd-tag 0.1.0 --httpd-image registry.opensuse.org/uyuni/proxy-httpd /path/to/config.tar.gz

```
It adjusts the configuration file for the httpd service, where
`registry.opensuse.org/uyuni/proxy-httpds` is the image to use and `0.1.0` is the version
tag, before restarting it.

要重置为默认值，请再次运行 install 命令但不要指定这些参数:
```

mgrpxy install /path/to/config.tar.gz

```
此命令首先将所有服务的配置重置为全局默认值，然后重新装载配置。

:leveloffset: 3
:leveloffset: +3

[[proxy-setup-containers-uyuni]]
= 容器化 {productname} Proxy 设置
:revdate: 2025-06-30
:page-revdate: {revdate}

为 {productname} Proxy 容器准备好容器主机后，需要额外执行几步容器设置才能完成配置。

.过程
. 生成 {productname} Proxy 配置归档文件
. 将配置归档传输到在安装步骤中准备的容器主机并解压缩
. Start the proxy sevices with `mgrpxy`


== 生成代理配置


{productname} 代理的配置归档由 {productname}
服务器生成。每个附加代理都需要自身的配置归档。

对于容器化 {productname}
代理，您必须构建新的代理配置文件，然后重新部署容器以使更改生效。此流程适用于更新设置（包括
SSL 证书）。

//[NOTE]
//====
//2 GB represents the default proxy squid cache size.
//This will need to be adjusted for your environment.
//====

[IMPORTANT]
====
对于 Podman 部署，在生成此代理配置之前，必须将 {productname}
代理的容器主机作为客户端注册到 {productname} 服务器。
====

If a proxy FQDN is used to generate a proxy container configuration that is not a
registered client (as in the Kubernetes use case), a new system entry will appear in
system list. This new entry will be shown under previously entered Proxy FQDN value and
will be of `Foreign` system type.

[NOTE]
====
Peripheral servers are always using third-party SSL certificates. If the hub server has
generated the certificates for the peripheral server, it needs to generate the certificate
of each proxy too.
```

```
On the hub server, run the following command.

[source, shell]
```

mgrctl exec -ti — rhn-ssl-tool --gen-server --dir="/root/ssl-build" --set-country="COUNTRY" \ --set-state="STATE" --set-city="CITY" --set-org="ORGANIZATION" \ --set-org-unit="ORGANIZATION UNIT" --set-email="name@example.com" \ --set-hostname=PROXY --set-cname="proxy.example.com"

```
需要使用的文件包括:

. `/root/ssl-build/RHN-ORG-TRUSTED-SSL-CERT` as the root CA,
. `/root/ssl-build/<hostname>/server.crt` as the proxy certificate and
. `/root/ssl-build/<hostname>/server.key` as the proxy certificate's key.
====

// tag::generate-proxy-config-section[]
=== 使用 {webui} 生成代理配置

.过程: 使用 {webui} 生成代理容器配置
[role=procedure]
-----

. 在 {webui} 中，导航到menu:系统[代理配置]，然后填写所需数据:

. In the `Proxy FQDN` field type fully qualified domain name for the proxy.

. In the `Parent FQDN` field type fully qualified domain name for the {productname} Server
or another {productname} Proxy.

. In the `Proxy SSH port` field type SSH port on which SSH service is listening on
{productname} Proxy. Recommended is to keep default 8022.

. In the `Max Squid cache size [MB]` field type maximal allowed size for Squid cache.
Recommended is to use at most 80% of available storage for the containers.

+

[NOTE]
====
2 GB 表示默认的代理 squid 缓存大小。需要根据您的环境调整此大小。
====

+

In the `SSL certificate` selection list choose if a new server certificate should be
generated for {productname} Proxy, an existing one should be used, or the SSL part should
be skipped entirely. You can consider generated certificates as {productname} builtin
(self signed) certificates. If {productname} server runs on {kube}, the generated
certificate option is not possible and replaced with no SSL certificate as they are
managed outside the containers.

+

Depending on the choice then provide either path to signing CA certificate to generate a
new certificate or path to an existing certificate and its key to be used as proxy
certificate. If the SSL part is skipped, the resulting configuration archive does not
carry `server_crt`, `server_key` or `ca_crt`; the deployment is then responsible for
providing the proxy certificate and root CA out-of-band (for {kube} deployments this means
creating `proxy-cert` and `uyuni-ca` manually — see the TLS setup section of the {kube}
proxy deployment page).
```

```
+

The CA certificates generated by the server are stored in the
`/var/lib/containers/storage/volumes/root/_data/ssl-build` directory.

+

有关现有或自定义证书的详细信息以及企业和中间证书的概念，请参见 xref:administration:ssl-
certs-imported.adoc[]。

. Click btn:[Generate] to register a new proxy FQDN in the {productname} Server and
generate a configuration archive (`config.tar.gz`) containing details for the container
host.

. 片刻之后，系统会显示文件可供下载。请将此文件保存在本地。

-----


=== Generate Proxy Configuration With `spacecmd` and Self-Signed Certificate

You can generate a Proxy configuration using `spacecmd`. This is only possible if
{productname} server runs on podman and has a self-signed root CA certificate.

.过程: 使用 spacecmd 和自我签名证书生成代理配置
[role=procedure]
-----
. 通过 SSH 连接到您的容器主机。

. 执行以下命令（替换其中的服务器和代理 FQDN）:

+
```

mgrctl exec -ti 'spacecmd proxy_container_config_generate_cert — dev-pxy.example.com dev-srv.example.com 2048 email@example.com -o /tmp/config.tar.gz'

```
. 从服务器容器复制生成的配置:

+
```

mgrctl cp server:/tmp/config.tar.gz

```
-----



=== Generate Proxy Configuration With `spacecmd` and Custom Certificate

You can generate a Proxy configuration using `spacecmd` for custom certificates rather
than the default self-signed certificates.

.过程: 使用 spacecmd 和自定义证书生成代理配置
[role=procedure]
-----
. 通过 SSH 连接到您的服务器容器主机。

. Execute the following commands, replacing the Server and Proxy FQDN:
```

```
+
```

for f in ca.crt proxy.crt proxy.key; do mgrctl cp $f server:/tmp/$f done mgrctl exec -ti 'spacecmd proxy_container_config — -p 8022 pxy.example.com srv.example.com 2048 email@example.com /tmp/ca.crt /tmp/proxy.crt /tmp/proxy.key -o /tmp/config.tar.gz'

```
+
. 如果您的设置使用中间 CA，请同时复制该证书，并在命令中通过 `-i`
选项（可根据需要多次提供）包含该证书：

+
```

mgrctl cp intermediateCA.pem server:/tmp/intermediateCA.pem mgrctl exec -ti 'spacecmd proxy_container_config — -p 8022 -i /tmp/intermediateCA.pem pxy.example.com srv.example.com 2048 email@example.com /tmp/ca.crt /tmp/proxy.crt /tmp/proxy.key -o /tmp/config.tar.gz'

```
. 从服务器容器复制生成的配置：
+
```

mgrctl cp server:/tmp/config.tar.gz

```
-----



=== Generate Proxy Configuration With `spacecmd` and no Certificate

You can generate a Proxy configuration using `spacecmd` with no TLS certificates. This is
needed for {productname} running on {kube} as the certificates are handled outside of the
containers.

.Procedure: Generating Proxy Configuration with spacecmd and no Certificate
[role=procedure]
-----
. 通过 SSH 连接到您的服务器容器主机。

. Execute the following commands, replacing the Server and Proxy FQDN:

+
```

for f in ca.crt proxy.crt proxy.key; do mgrctl cp $f server:/tmp/$f done mgrctl exec -ti 'spacecmd proxy_container_config_nossl — -p 8022 pxy.example.com srv.example.com 2048 email@example.com -o /tmp/config.tar.gz'

```
. 从服务器容器复制生成的配置：
+
```

mgrctl cp server:/tmp/config.tar.gz

```
-----

// end::generate-proxy-config-section[]
```

```
[[proxy-setup-containers-transfer-config]]
== 传输 {productname} 代理配置


Both `spacecmd` command and generating via {webui} ways create a configuration archive.
This archive needs to be made available on container host. Transfer this generated archive
to the container host.



[[proxy-setup-containers-transfer-start]]
== 启动 {productname} 代理容器

Container can be started with the `mgrpxy` command.

[[proc-setup-containers-setup-start]]
.过程：启动 {productname} 代理容器

. 运行以下命令：
+
```

mgrpxy start uyuni-proxy-pod

```
+
. 调用以下命令检查所有容器是否都已按预期启动：
+
```

podman ps

```
Five {productname} Proxy containers should be present and should be part of `proxy-pod`
container pod.

* proxy-salt-broker
* proxy-httpd
* proxy-tftpd
* proxy-squid
* proxy-ssh

:leveloffset: 3
:leveloffset: +3

[[proxy-conversion-from-client-uyuni]]
= 从客户端转换为代理

== 概述

本章介绍如何通过 {webui} 将客户端系统转换为 {productname} 代理。

本文假定代理主机系统已完成引导，且已订阅基础操作系统通道。

有关客户端初始配置的详细信息，请参见 xref:client-configuration:registration-
overview.adoc[]。

== 要求

开始转换前，请确保满足以下要求。


=== 客户端需满足的条件
```

```
- 已在 {productname} 中完成初始配置
- 可通过网络访问


== 准备工作

进行代理转换前，请完成以下准备工作，以免转换过程中出现中断。

=== SSL 证书

为确保代理与其他组件之间的通信安全，需要具备有效的 SSL 证书。

您需要准备：

* 为 {productname} 服务器证书签名的证书颁发机构 (CA) 的公共证书。
* 代理专用证书。
* 代理证书对应的私用密钥。

[NOTE]
====
如果您的 CA 使用中间证书链，还必须包含所有中间证书。
====

如果不使用第三方证书，可在 {productname} 容器内通过 `rhn-ssl-tool` 生成证书。

.生成代理证书

. 在 {productname} 服务器主机上运行以下命令：
+
[source, shell]
```

mgrctl exec -ti — rhn-ssl-tool --gen-server \ --set-hostname="<PROXY-FQDN>" \ --dir="/root/ssl-build"

```
+
有关其他参数的详细信息，请参见 xref:administration:ssl-certs-selfsigned.adoc[]。
+

. 将证书传输至 {productname} 服务器主机
+
[source, shell]
```

mgrctl cp server:/root/ssl-build/<PROXY-FQDN>/server.crt /root/proxycert.pem mgrctl cp server:/root/ssl-build/<PROXY-FQDN>/server.key /root/proxykey.pem mgrctl cp server:/root/ssl-build/RHN-ORG-TRUSTED-SSL-CERT /root/rootca.pem

```
+
[NOTE]
====
如需确认证书和密钥文件的具体生成目录，可以执行以下命令列出相关目录：
```

mgrctl exec -ti — ls -ltd /root/ssl-build/*/

```
====

. 从 {productname} 服务器主机传输证书
+
```

```
[source, shell]
```

scp <UYUNI-FQDN>:/root/proxycert.pem ./ scp <UYUNI-FQDN>:/root/proxykey.pem ./ scp <UYUNI-FQDN>:/root/rootca.pem ./

```
=== 软件包准备

==== 安装 `mgrpxy` 工具

必须从与您的系统匹配的储存库安装 `mgrpxy` 工具。请从以下地址选择适当的储存库：

https://download.opensuse.org/repositories/systemsmanagement:/Uyuni:/Stable:/ContainerUtil
s/

.Example {opensuse} {tumbleweed} installation:
[source, shell]
```

zypper ar https://download.opensuse.org/repositories/systemsmanagement:/Uyuni:/Stable:/ContainerUtils/openSUSE_Tumbleweed/ uyuni-containerutils zypper ref zypper in mgrpxy

```
==== 安装容器映像

建议以 RPM 软件包形式部署容器映像。请确保客户端上已安装以下软件包：

[source, shell]
```

zypper ar https://download.opensuse.org/repositories/systemsmanagement:/Uyuni:/Stable/containerfile/ uyuni-proxy-images zypper ref zypper in uyuni-proxy-httpd-image \ uyuni-proxy-salt-broker-image \ uyuni-proxy-squid-image \ uyuni-proxy-ssh-image \ uyuni-proxy-tftpd-image

```
有关隔离部署的详细信息，请参见 xref:installation-and-upgrade:container-
deployment/mlm/proxy-air-gapped-deployment-mlm.adoc[]。


== 设置代理客户端

. Navigate to the client's `Overview` page.
. 单击 btn:[转换为代理] 按钮。
+
确认您已重定向至代理配置表单。
+
This page can be accessed later from the `Details` > `Proxy` > `Configuration` tab.

. 在 {webui} 中，导航到menu:代理[配置]，并填写所需数据：

+
.过程：配置代理
.. In the `Parent FQDN` field, type the fully qualified domain name for the parent server
or proxy.
.. In the `Proxy SSH port` field, type the SSH port on which the SSH service is listening
on the {productname} Proxy. It is recommended to keep the default: 8022.
.. In the `Max Squid cache size` field, type the maximum allowed size for the Squid cache,
in Gigabytes.
.. In the `Proxy admin email` field, type the administrator's email address.
.. In the `Certificates` section, provide the certificates for the {productname} Proxy,
obtained in the preparation step.
```

```
.. In the `Source` section, select one of the two options: `RPM` or `Registry`.
+
* The `RPM` option is recommended for air-gapped or restricted environments.
The `Registry` option can be used if connectivity to the container image registry is
available. + If selected, you will be prompted to choose between two sub-options: `Simple`
or `Advanced`.
+
** If `Simple` is selected, provide values in the `Registry URL` and `Containers Tag`
fields.
*** For `Registry URL` use: `registry.opensuse.org/uyuni`.
*** 从下拉列表中选择标记。

** If `Advanced` is selected, an additional section of the form is shown:
*** For each individual container URL field, use the registry:
`registry.opensuse.org/uyuni` followed by the corresponding suffix, for example, `_proxy-
httpd_` or `_salt-broker_`.
*** 从下拉列表中选择标记。

. 填写完所有字段后，单击 btn:[应用] 按钮保存配置并调度代理安装任务。

== 校验代理激活状态

查看客户端的事件历史记录，确认任务执行成功。

（可选）访问代理的 HTTP 端点，验证其是否显示欢迎页面。

:leveloffset: 3


== 升级和迁移
=== 服务器
:leveloffset: +3

= Migrating the {productname} Server to {opensuse} Tumbleweed
:revdate: 2025-10-27
:page-revdate: {revdate}
:description: This page describes how to migrate a {productname} Server running on
openSUSE Leap Micro 5.5 to a fresh host running openSUSE Tumbleweed as the base OS.

This page describes a simple, backup-and-restore migration of a {productname} Server
running on openSUSE Leap Micro 5.5 to a fresh host running openSUSE Tumbleweed as the base
OS.


== Overview of the Migration Process

You will:

* Create a full server backup with `mgradm backup` on the openSUSE Leap Micro 5.5 host.
* Reinstall the host with openSUSE Tumbleweed (server profile).
* Install {productname} tools and prerequisites on Tumbleweed.
* Restore the backup with `mgradm backup restore`.
* Start services and verify the server.


== 要求和注意事项

* Source server: openSUSE Leap Micro 5.5 running {productname} (for example:
{productnumber}).
* Target server: openSUSE Tumbleweed with the same hostname/FQDN and IP (recommended) to
avoid client-side changes.
* SSH/scp access between machines for transferring the backup tarball.
* Sufficient free disk space on both source and target for the backup and restore.

[IMPORTANT]
```

```
====
Restore to the same {productname} version you backed up, or a version explicitly
documented as compatible for restore. If you use development or preview repositories (for
example, Uyuni Master), expect changes and re-validate.
====

== Migration Procedure

=== Step 1: Create a Backup on the openSUSE Leap Micro 5.5 Server
.Procedure:  Create a Backup
[role="procedure"]
____
. As root on the old server, create a backup directory and run the backup:

+

[source, shell]
```

mgradm backup /tmp/uyuni-backup

```
. Package the backup for transfer:

+

[source, shell]
```

tar -C /tmp -cvf /tmp/uyuni-backup.tar uyuni-backup

```
. Copy the backup to a safe location you can reach from the new host:

+

[source, shell]
```

scp /tmp/uyuni-backup.tar <USER>@<HOST>:/path/to/store/

```
[TIP]
====
You can store the backup to external storage or an object store as long as you can fetch
it on the new host.
====


----

=== Step 2: Reinstall the Host with openSUSE Tumbleweed
.Procedure: Reinstalling the Host
[role="procedure"]
----

. Reprovision the VM or bare-metal host with openSUSE Tumbleweed.
. Choose a basic  "server profile"  installation.
. Set the same hostname/FQDN and IP address as the original server if you want clients to
reconnect seamlessly.


----

=== Step 3: Install {productname} Tools and Prerequisites on Tumbleweed

.Procedure: Installing Tools and Prerequisites
```

```
[role="procedure"]
----

. Add the Uyuni Stable repository and install tools:
+
[source, shell]
```

zypper ar https://download.opensuse.org/repositories/systemsmanagement:/Uyuni:/Stable/images/repo/Uyuni-Server-POOL-x86_64-Media1 uyuni-server-stable zypper ref zypper in mgradm mgrctl mgradm-bash-completion mgrctl-bash-completion uyuni-storage-setup-server

```
. Install Podman if it was not automatically pulled in:

+

[source, shell]
```

zypper in podman

```
[NOTE]
====
The package `uyuni-storage-setup-server` provides the `mgr-storage-server` tool for
preparing persistent volumes. Installing `podman` explicitly may be necessary on some
installations.
====


----

=== Step 4: Optional - Prepare Persistent Storage

.Procedure: Prepareing Persistent Storage
[role="procedure"]
----

It is recommended to configure persistent storage with `mgr-storage-server` to avoid
container full-disk issues.

[source, shell]
```

mgr-storage-server <storage-disk-device> [<database-disk-device>]

```
[NOTE]
====
Devices must be raw (no existing filesystem). The tool creates volumes at
`/var/lib/containers/storage/volumes`.

For details, see:

* xref:installation-and-upgrade:container-management/persistent-container-volumes.adoc[]
* xref:administration:troubleshooting/tshoot-container-full-disk.adoc[]
====


----

=== Step 5 Fetch and Restore the Backup on Tumbleweed

.Procedure: Fetching and Restoring the Backup
[role="procedure"]
```

```
----
. Copy the backup to the new server and unpack it:
+
[source, shell]
```

scp <USER>@<HOST>:/path/to/store/uyuni-backup.tar /tmp/ tar -C /tmp -xvf /tmp/uyuni-backup.tar

```
. Restore using `mgradm` (point to the extracted backup directory):
+
[source, shell]
```

mgradm backup restore /tmp/uyuni-backup

```
----
=== Step 6: Start Services and Verify
.Procedure: Starting Services and Verifying
. Start the server services:
+
[source, shell]
```

mgradm start

```
. Verify:
** Check that all containers are up: `mgrctl ps` or `podman ps`.
** Access the Web UI (HTTPS) and log in.
** Review logs for errors: `mgrctl logs server` and other components as needed.

** ___

== Notes and Troubleshooting

* If Podman wasn’ t installed automatically, install it with `zypper in podman` and rerun
the restore/start steps.
* Ensure the target host has the same time, hostname, and IP configuration expected by
your setup (especially if clients exist).
* For large environments, ensure adequate disk throughput and space. The backup and
restore can take a long time.

[IMPORTANT]
====
If the restore fails or the new system cannot start, you can still boot the original
openSUSE Leap Micro 5.5 system and continue service. Keep the original VM/snapshots until
you fully validate the new Tumbleweed-based server.
====


:leveloffset: 3
=== 代理
:leveloffset: +3
```

```
= Migrating the {productname} Proxy to {opensuse} Tumbleweed
:revdate: 2025-10-27
:page-revdate: {revdate}
:description: This page describes how to migrate a {productname} Proxy host from openSUSE
Leap Micro 5.5 to a fresh openSUSE Tumbleweed installation using the proxy administration
tool `mgrpxy`.

This page describes how to migrate a {productname} Proxy host from openSUSE Leap Micro 5.5
to a fresh openSUSE Tumbleweed installation using the proxy administration tool `mgrpxy`.

[IMPORTANT]
====
This guide was tested on Tumbleweed only. There is no known reason it wouldn’ t work on
other supported bases, but always validate in a test environment before production.
====



== Overview of the Proxy Migration Process

You will:

* Save proxy configuration from the old system (including Apache/Squid tuning).
* Reinstall the host with openSUSE Tumbleweed.
* Re-register the host using the system reactivation key.
* Install `mgrpxy` (and Podman if needed).
* Restore configuration and run `mgrpxy install` with optional tuning files.


== 要求和注意事项

* Keep the same hostname/FQDN and IP when possible so the server and clients interact with
the proxy as before.
* Ensure you have the “system reactivation key” for the existing proxy system (UI:
Systems > select the proxy > Details > Reactivation).
* Ensure SSH/scp access to move configuration archives off and onto the machine.

== Migration Procedure

=== Step 1: Save Proxy Configuration and Tuning Files

.Procedure: Save Proxy Configuration and Tuning Files
[role="procedure"]
----

. Copy the Uyuni proxy configuration directory to a safe location:

+

[source, shell]
```

scp -r /etc/uyuni <USER>@<HOST>:/some/where/safe/

```
. Identify Apache and Squid tuning files currently in use by the legacy proxy services:

+

[source, shell]
```

systemctl cat uyuni-proxy-httpd.service | grep EXTRA_CONF= | sed 's/.**=-v\([^:]+\):.**/\1/' systemctl cat uyuni-proxy-squid.service | grep EXTRA_CONF= | sed 's/.**=-v\\([^:]\+\\):.**/\1/'

```
. Copy those tuning files to the same safe location as well.

[TIP]
====
Typical default paths after you copy them back will be:

* Apache tuning: `/etc/uyuni/proxy/apache.conf`
* Squid tuning: `/etc/uyuni/proxy/squid.conf`
====


----

=== Step 2: Reinstall the Host with openSUSE Tumbleweed
.Procedure: Reinstalling the Host with openSUSE Tumbleweed
[role="procedure"]
----

. Reinstall the machine with openSUSE Tumbleweed (server profile recommended).
. Set the same hostname/FQDN and IP as before when possible.


----



=== Step 3: Re-register the Host with the Reactivation Key

.Procedure: Re-registering the Host with the Reactivation Key
[role="procedure"]
----

. From the {productname} Web UI, obtain the system reactivation key for the existing proxy
system record (Systems > Details > Reactivation).
. Bootstrap/re-register the Tumbleweed host using that reactivation key so it claims the
existing system entry.

[NOTE]
====
Use your standard bootstrapping process for Tumbleweed hosts in your environment (for
example, the bootstrap script or your configuration management), ensuring the reactivation
key is applied.
====


----



=== Step 4: Install {productname} Proxy Tools and Podman

.Procedure: Installing Proxy Tools and Podman
[role="procedure"]
----

. Add the Uyuni Stable repository and install tools:

+

[source, shell]
```

zypper ar https://download.opensuse.org/repositories/systemsmanagement:/Uyuni:/Stable/images/repo/Uyuni-Proxy-POOL-x86_64-Media1 uyuni-proxy-stable zypper ref zypper in mgrpxy mgrctl mgrpxy-bash-completion mgrctl-bash-completion

```
. Ensure Podman is installed (required to run containers):

+

[source, shell]
```

zypper in podman

```
----



=== Step 5: Restore Configuration and Install the Proxy

.Procedure: Restoring Configuration and Install the Proxy
[role="procedure"]
----

. Copy back the saved configuration directory to the new host:

+

[source, shell]
```

scp -r <USER>@<HOST>:/some/where/safe/uyuni /etc/

```
. If you saved Apache/Squid tuning files, place them at the expected default paths or note
their locations for parameters in the next command:

+

[source, shell]
```

3.1.2. Default paths expected by mgrpxy parameters (adjust/move your files accordingly)

3.1.3. Apache tuning: /etc/uyuni/proxy/apache.conf

3.1.4. Squid tuning: /etc/uyuni/proxy/squid.conf

```
. Run the proxy installation with Podman. If you do not use tuning files, omit the
corresponding parameters:

+

[source, shell]
```

3.1.5. With tuning files

mgrpxy install \ --tuning-httpd /etc/uyuni/proxy/apache.conf \ --tuning-squid /etc/uyuni/proxy/squid.conf

3.1.6. If you have no tuning files, remove the tuning parameters:

3.1.7. mgrpxy install

```
[NOTE]
====
In an upcoming release, if tuning files are placed at the default paths noted above, the
explicit parameters will not be required.
====


----


=== Step 6: Verify the Proxy

.Procedure: Verifying the Proxy
[role="procedure"]
----

. Check containers are running:

+

[source, shell]
```

mgrctl ps # or podman ps

```
. Confirm the proxy appears healthy in the {productname} Web UI and that clients using
this proxy operate normally.


----

== 查错

* If Podman was missing, install it and rerun the `mgrpxy install` step.
* Verify the host’ s time, hostname, and IP match expectations.
* If the host did not reattach to the existing system record, confirm you used the correct
reactivation key and repeat the bootstrap.


:leveloffset: 3
:leveloffset: +3

= 将旧版代理迁移到容器
:revdate: 2025-02-06
:page-revdate: {revdate}

The containerized proxy now is managed by a set of systemd services. For managing the
containerized proxy, use the `mgrpxy` tool.

This section will help you migrate from the legacy `systemd` proxy using the `mgrpxy`
tool.

[IMPORTANT]
====
由于主机操作系统已从 {leap} 更改为 {leapmicro}，因此不支持从旧版 {productname} 就地迁移到
{productnumber}。

{productname} {productnumber} 及更高版本不再支持传统联系协议。从旧版 {productname} 迁移到
{productnumber} 之前，必须将所有现有的传统客户端（包括传统代理）迁移到 {salt}。
====
```

```
== 使用 Systemd 从旧版迁移到容器化代理


=== 生成代理配置

.过程: 生成代理配置
. 登录到 {productname} 服务器 {webui}。
. 在左侧导航栏中，选择menu:系统[代理配置]。
. 输入您的代理 FQDN。使用与原始代理主机相同的 FQDN。
. 输入您的服务器 FQDN。
. 输入代理端口号。__建议使用默认端口 8022。__
. 证书和私用密钥位于服务器容器主机上的
`/var/lib/containers/storage/volumes/root/_data/ssl-build/` 中。
  * RHN-ORG-TRUSTED-SSL-CERT
  * RHN-ORG-PRIVATE-SSL-KEY
. 使用以下命令将证书和密钥复制到您的计算机:
+
```

scp root@uyuni-server-example.com:/root/ssl-build/RHN-ORG-PRIVATE-SSL-KEY scp root@uyuni-server-example.com:/root/ssl-build/RHN-ORG-TRUSTED-SSL-CERT

```
. 选择 btn:[选择文件] 并在本地计算机上通过浏览找到证书。
. 选择 btn:[选择文件] 并在本地计算机上通过浏览找到私用密钥。
. 输入 CA 口令。
. 单击 btn:[生成]。


=== 将代理配置传输到新主机

.过程: 传输代理配置
. 在服务器中，将生成的包含代理配置的 tar.gz 文件传输到新代理主机:
+
```

scp config.tar.gz <uyuni 代理 FQDN>:/root/

```
. 在执行下一步之前，请先禁用旧版代理:
+
```

spacewalk-proxy stop

```
. 使用以下命令部署新代理:
+
```

systemctl start uyuni-proxy-pod

```
. 使用以下命令启用新代理:
+
```

systemctl enable --now uyuni-proxy-pod

```
. 运行 ``podman ps`` 来校验所有容器是否存在并正在运行:
```

```
+
```

proxy-salt-broker proxy-httpd proxy-tftpd proxy-squid proxy-ssh

```
== 将 {productname} 代理迁移到 {productname} {productnumber} 容器化代理

.过程: 将 {productname} 容器化代理迁移到 {productname} {productnumber} 新容器化代理
. 引导新计算机，然后开始安装 {leapmicro} {microversion}。
. 完成安装。
. 更新系统:
+
```

transactional-update --continue

```
. Install `mgrpxy` and optionally, `mgrpxy-bash-completion`:
+
```

transactional-update pkg install mgrpxy mgrpxy-bash-completion

```
+
. 重引导。

. Copy your `tar.gz` proxy configuration to the host.



== 使用 {webui} 安装软件包

The `mgrpxy` and `mgrpxy-bash-completion` packages can also be installed via the web UI
after the minion has been bootstrapped and registered with the Server.

.过程: 使用 {webui} 安装软件包
. 安装后，确保在menu:管理[安装向导 -> 产品]页面中添加并同步 {sle-micro} {microversion}
父通道和代理子通道。
. 在 {webui} 中，转到menu:系统[激活密钥]，并创建一个与已同步 {sle-micro} {microversion}
通道关联的激活密钥。
. 使用menu:系统[引导]页面将系统作为受控端进行引导。
. 在初始配置新计算机并且其显示在系统列表中后，选择系统并导航到menu:系统细节[安装软件包
]页面。
. Install the packages `mgrpxy` and `mgrpxy-bash-completion`.
. 重引导系统。


== Generate Proxy Config With `spacecmd` and Self-Signed Certificate

可以使用 spacecmd 生成代理配置。

.Procedure: Generate Proxy Config With `spacecmd` and Self-Signed Certificate
. 通过 SSH 连接到您的容器主机。
. 执行以下命令（替换其中的服务器和代理 FQDN）:
+
```

mgrctl exec -ti 'spacecmd proxy_container_config_generate_cert — dev-pxy.example.com dev-srv.example.com 2048 email@example.com -o /tmp/config.tar.gz'

```
. 将生成的配置复制到代理:
```

```
+
```

mgrctl cp server:/tmp/config.tar.gz

```
. 使用以下命令部署代理:
+
```

mgrpxy install config.tar.gz

```
== Generate Proxy Config With `spacecmd` and Custom Certificate

You can generate Proxy configuration using `spacecmd` for a custom certificates rather
than default self-signed certificates.

[NOTE]
====
2 GB 表示默认的代理 squid 缓存大小。需要根据您的环境调整此大小。
====

.Procedure: Generate Proxy Config With `spacecmd` and Custom Certificate
. 通过 SSH 连接到您的服务器容器主机。
. 执行以下命令（替换其中的服务器和代理 FQDN）：
+
```

for f in ca.crt proxy.crt proxy.key; do mgrctl cp $f server:/tmp/$f done mgrctl exec -ti 'spacecmd proxy_container_config -- -p 8022 pxy.example.com srv.example.com 2048 email@example.com /tmp/ca.crt /tmp/proxy.crt /tmp/proxy.key -o /tmp/config.tar.gz'

```
. 将生成的配置复制到代理:
+
```

mgrctl cp server:/tmp/config.tar.gz

```
. 使用以下命令部署代理:
+
```

mgrpxy install config.tar.gz

```
:leveloffset: 3
=== 客户端
:leveloffset: +3

[[client-upgrade]]
= 升级客户端
:revdate: 2024-11-19
:page-revdate: {revdate}


客户端采用底层操作系统的版本控制系统。对于运行 {suse} 操作系统的客户端，可在 {productname}
{webui} 中进行升级。

有关升级客户端的详细信息，请参见 xref:client-configuration:client-upgrades.adoc[]。

:leveloffset: 3
```

```
== 基本的服务器和代理管理
:leveloffset: +2

= 使用 mgradm 进行自定义 YAML 配置和部署
:revdate: 2024-11-19
:page-revdate: {revdate}

You have the option to create a custom `mgradm.yaml` file, which the `mgradm` tool can
utilize during deployment. All `mgradm` arguments have their YAML counterparts. For
example, the `mgradm` argument `--ssl-db-ca-intermediate` can be specified in the
`mgradm.yaml` file as follows:

[source, yaml]
ssl:
  db:
    ca:
      intermediate: /path/to/ca-intermediate.crt

[IMPORTANT]
====
`mgradm` will prompt for basic variables if they are not provided using command line
parameters or the `mgradm.yaml` configuration file.

为了安全起见，**应避免使用命令行参数指定口令**。请改用具有适当权限的配置文件。
====

.过程：使用自定义配置文件通过 Podman 部署 {productname} 容器
. Prepare a configuration file named `mgradm.yaml` similar to the following example:
+
[source, yaml]
....
# 数据库口令。默认会随机生成
db:
  password: MySuperSecretDBPass

# CA 证书的口令
ssl:
  password: MySuperSecretSSLPassword

# 您的 SUSE Customer Center 身份凭证
scc:
  user: ccUsername
  password: ccPassword

# 组织名称
organization: YourOrganization

# 用于发送通知的电子邮件地址
emailFrom: notifications@example.com

# 管理员帐户细节
admin:
  password: MySuperSecretAdminPass
  login: LoginName
  firstName: Admin
  lastName: Admin
  email: email@example.com
....

. 在终端中，以 root 身份运行以下命令。服务器 FQDN 是选填的。
+
[source, shell]
```

mgradm -c mgradm.yaml install podman <FQDN>

```
+
[IMPORTANT]
====
必须以 sudo 或 root 用户身份部署容器。如果您遗漏此步骤，终端中将显示以下错误。

[source, shell]
```

INF Setting up uyuni network 9:58AM INF Enabling system service 9:58AM FTL Failed to open /etc/systemd/system/uyuni-server.service for writing error="open /etc/systemd/system/uyuni-server.service: permission denied"

```
====

. 等待部署完成。

. 打开浏览器并访问您的服务器 FQDN 或 IP 地址。

//In this section you learned how to deploy an {productname} {productnumber} Server
container using a custom YAML configuration.

:leveloffset: 3
:leveloffset: +2

= 启动和停止容器
:revdate: 2024-02-13
:page-revdate: {revdate}

可使用以下命令重启、启动和停止 {productname} {productnumber} 服务器容器：


To `restart` the {productname} {productnumber} Server execute the following command:
```

3.1.8. mgradm restart

5:23PM INF Welcome to mgradm 5:23PM INF Executing command: restart

```
To `start` the server execute the following command:
```

3.1.9. mgradm start

5:21PM INF Welcome to mgradm 5:21PM INF Executing command: start

```
To `stop` the server execute the following command:
```

3.1.10. mgradm stop

5:21PM INF Welcome to mgradm 5:21PM INF Executing command: stop

```
// Coming soon:
//You can also check on the status of services running in the container with:
```

```
//----
//mgradm status
//----

:leveloffset: 3
:leveloffset: +2

[[container-list]]
= {productname} 使用的容器
:revdate: 2025-05-08
:page-revdate: {revdate}

下面列出了 {productname} {productnumber} 使用的容器。

.服务器容器
[cols="name,description"]
|===
|容器名称 | 说明

| uyuni-server
| 主产品容器

| uyuni-db
| 产品的数据库容器

| uyuni-hub-xmlrpc
| 用于 Hub 部署的 XML-RPC 网关

| uyuni-server-attestation
| 服务器 COCO 证明容器

| uyuni-saline
| 用于实现 Salt 可观测性的 Saline 容器

| uyuni-server-migration
| 迁移辅助容器
|===

.代理容器
[cols="name,description"]
|===
|容器名称 | 说明

| uyuni-proxy-httpd
| 处理所有 HTTP 通信的主代理容器

| uyuni-proxy-squid
| Squid 缓存容器

| uyuni-proxy-salt-broker
| Salt 转发器容器

| uyuni-proxy-ssh
| SSH 转发器容器

| uyuni-proxy-tftpd
| TFTP 到 HTTP 的转换器与转发器容器
|===

:leveloffset: 3
:leveloffset: +2

[[persistant-volume-list]]
= 永久性容器卷
:revdate: 2025-07-29
```

```
:page-revdate: {revdate}

在容器中执行的修改不会保留。在永久性卷外部所做的任何更改都将被丢弃。下面列出了
{productname} {productnumber} 的永久性卷。

To customize the default volume locations, ensure you create the necessary volumes before
launching the pod for the first time, utilizing the `podman volume create` command.

[NOTE]
====
请确保此表格与 Helm 图表和 systemctl 服务定义中所述的卷映射完全一致。
====


== 服务器


以下卷存储在服务器上的 **Podman** 默认存储位置。

.永久性卷: **Podman 默认存储**
[cols="name,directory"]
|===
|Volume Name | Volume Directory

| **Podman Storage**
| `/var/lib/containers/storage/volumes/`
|===

.永久性卷: **root**
[cols="name,directory"]
|===
|Volume Name | Volume Directory

| **root**
| `/root`
|===


.永久性卷: **var/**
[cols="name,directory"]
|===
|Volume Name | Volume Directory

| **var-cobbler**
| `/var/lib/cobbler`

| **var-salt**
| `/var/lib/salt`

| **var-pgsql**
| `/var/lib/pgsql/data`

| **var-pgsql-backup**
| `/var/lib/pgsql-backup`

| **var-cache**
| `/var/cache`

| **var-spacewalk**
| `/var/spacewalk`

| **var-log**
| `/var/log`
|===
```

```
.永久性卷: **srv/**
[cols="name,directory"]
|===
| Volume Name | Volume Directory

| **srv-salt**
| `/srv/salt`

| **srv-www**
| `/srv/www/`

| **srv-tftpboot**
| `/srv/tftpboot`

| **srv-formulametadata**
| `/srv/formula_metadata`

| **srv-pillar**
| `/srv/pillar`

| **srv-susemanager**
| `/srv/susemanager`

| **srv-spacewalk**
| `/srv/spacewalk`
|===



.永久性卷: **etc/**
[cols="name,directory"]
|===
|Volume Name | Volume Directory

| **etc-apache2**
| `/etc/apache2`

| **etc-rhn**
| `/etc/rhn`

| **etc-systemd-multi**
| `/etc/systemd/system/multi-user.target.wants`

| **etc-systemd-sockets**
| `/etc/systemd/system/sockets.target.wants`

| **etc-salt**
| `/etc/salt`

| **etc-sssd**
| `/etc/sssd`

| **etc-tomcat**
| `/etc/tomcat`

| **etc-cobbler**
| `/etc/cobbler`

| **etc-sysconfig**
| `/etc/sysconfig`

| **etc-postfix**
| `/etc/postfix`
```

```
| **ca-cert**
| `/etc/pki/trust/anchors`

|===


.永久性卷: **run/**
[cols="name,directory"]
|===
|Volume Name | Volume Directory

| **run-salt-master**
| `/run/salt/master`

|===



== 代理

以下卷存储在代理上的 **Podman** 默认存储位置。

.永久性卷: **Podman 默认存储**
[cols="name,directory"]
|===
|Volume Name | Volume Directory

| **Podman Storage**
| `/var/lib/containers/storage/volumes/`
|===

.永久性卷: **srv/**
[cols="name,directory"]
|===
|Volume Name | Volume Directory

| **uyuni-proxy-tftpboot**
| `/srv/tftpboot`
|===


.永久性卷: **var/**
[cols="name,directory"]
|===
|Volume Name | Volume Directory

| **uyuni-proxy-rhn-cache**
| `/var/cache/rhn`


| **uyuni-proxy-squid-cache**
| `/var/cache/squid`
|===

:leveloffset: 3
:leveloffset: +2

[[understanding-storage-scripts]]
= Understanding `mgr-storage-server` and `mgr-storage-proxy`
:revdate: 2025-07-07
:page-revdate: {revdate}
```

```
`mgr-storage-server` and `mgr-storage-proxy` are helper scripts provided with
{productname}.

它们用于为 {productname} 服务器和代理指定存储配置。

The scripts take disk devices as arguments. `mgr-storage-proxy` requires a single argument
for the storage disk device. `mgr-storage-server` requires a storage disk device and can
optionally accept a second argument for a dedicated database disk device. While both
normal and database storage can reside on the same disk, it is advisable to place the
database on a dedicated, high-performance disk to ensure better performance and easier
management.


== 这些工具的功能


Both `mgr-storage-server` and `mgr-storage-proxy` perform standard storage setup
operations:

* 验证提供的存储设备。
* 确保设备为空且适合使用。
* 在指定设备上创建 XFS 文件系统。
* 临时挂载设备以进行数据迁移。
* 将相关存储目录迁移到新设备。
* Create entries in `/etc/fstab` so that the storage mounts automatically on boot.
* 将设备重新挂载到最终位置。


.工具的专属特性
[cols="1,3a"]
|===
| `mgr-storage-server`
|
Optionally supports a separate device for database storage.

Stops {productname} services during migration, restarts them afterward.

Moves Podman volumes directory `/var/lib/containers/storage/volumes` to the prepared
storage, and optionally `/var/lib/containers/storage/volumes/var-pgsql` to the prepared
database storage.

| `mgr-storage-proxy`
|
Focuses only on proxy storage (no database storage support).

Stops and restarts the proxy service during migration.

Moves podman volumes directory `/var/lib/containers/storage/volumes` to the prepared
storage.

|===


[NOTE]
====
这两个工具均会自动执行标准的 Linux 存储操作，不存在 Linux
管理员手动操作之外的隐藏逻辑或自定义逻辑。
====


== 这些工具**不**具备的功能

* **不会**创建或管理 LVM 卷。
* **不会**配置 RAID 或复杂存储拓扑。
* **不会**阻止您在完成设置后使用常规 Linux 工具管理存储设备。
```

```
* **不**提供动态调整大小或扩展功能 — 这些操作必须通过标准 Linux 存储工具完成。


== 安装后的存储管理

完成存储配置后，您可通过标准 Linux 命令安全地管理存储设备。


=== 示例

.示例 1：扩展存储容量（如果使用 LVM）
```

lvextend -L +10G /dev/your_vg/your_lv xfs_growfs /var/lib/containers/storage/volumes

```
.示例 2：迁移到更大容量的磁盘

. 添加并格式化新磁盘。
. 临时挂载新磁盘。
. 使用 `rsync` 复制数据。
. 更新 `/etc/fstab`。
. 将新磁盘重新挂载到适当位置。


== 适用与不适用场景

[IMPORTANT]
====
更改存储设置前，务必创建备份。
====

* **仅**在初始存储设置或迁移到新存储设备（需要使用工具处理数据迁移和更新
`/etc/fstab`）时使用这些工具。
* Do *not* rerun these scripts for resizing or expanding storage. Use standard Linux tools
(e.g., `lvextend`, `xfs_growfs`) for such operations.


== 总结

`mgr-storage-server` and `mgr-storage-proxy` help automate the initial persistent storage
setup for {productname} components using standard Linux storage practices. They do not
limit or interfere with standard storage management afterward.

完成设置后，您可继续使用熟悉的 Linux 工具管理存储设备。

[IMPORTANT]
====
A full database volume can cause significant issues with system operation. As disk usage
notifications have not yet been adapted for containerized environments, users are
encouraged to monitor the disk space used by Podman volumes themselves, either through
tools such as Grafana, Prometheus, or any other preferred method. Pay particular attention
to the var-pgsql volume, located under `/var/lib/containers/storage/volumes/`.
====

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:leveloffset: +1

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